The a_0 Law as a Measurement, Form B and Its Rules, the Reach–Amount Lock, and the Symmetry That Makes κ a Convention

Working note (PAPER74), version 1.0, 2026-11-11. AI-assisted research programme; not peer reviewed; not reproduced by any outside group. Companions: PAPER69 (DOI 10.5281/zenodo.24025170, the closure record), PAPER70 (DOI 10.5281/zenodo.24092073, field equations), PAPER71 (DOI 10.5281/zenodo.24095262, the switch class), PAPER72 (v1.1 DOI 10.5281/zenodo.24125791, the form the record closes on) and PAPER73 (the lock). $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. The cold component is a gravitating field component, never a particle; it is never "cold dark matter". Nothing here says the data favour this framework.

Abstract

The non-baryonic mass in and around galaxies is real, conserved and gravitating, and there is no particle in it. What the data measure about it is one relation, and that relation is a measurement, not a hypothesis: the enclosed non-baryonic-to-baryonic mass ratio follows $X(x) = \frac{1}{2}(\sqrt{1+4x^2} - 1)$ with $x = r/r_M$, to a worst bin median of 0.057 dex canonical and 0.021 dex alt over $x = 0.18$ to 10 against a pre-registered line of 0.10 (SHP1, 2700 points, 149 galaxies); on host-free KiDS lenses the measured outer amplitude is $1.4002 \pm 0.2211/1.2740 \pm 0.2012$ of the law against a model of 1.003 and a pre-registered acceptance line of 0.891 (ONE1, reproduced by GAS1). Form B — the a_0 law with cold dark energy counted once — is this component's phenomenology, stated here as six rules, each tagged certified, hand-checked or measured. The week's negative result is then a lock, not a list: two exact theorems cap a halo at the charge the clock carried in. T4: inside any ball whose boundary free-falls the net polarization charge is exactly zero at all times, for any momentum equation, so charge cannot be manufactured during infall. LFS1: for a standing clock the polarization is frozen, the charge moving only by order $|\Phi|/c^2 \sim 10^{-6}$. Against a cosmic budget of $5.41\,M_b$ that is already spent at 38.2/34.7 kpc, the law demands $41.5\,M_b$ at the inner edge of KiDS's window and $97.8\,M_b$ at its middle edge; the demand over budget is 10.99–11.01 across four KiDS masses and both footings, i.e. mass- and footing-independent. The one flank that could have paid it — missing baryons in $g_{\rm bar}$ — is closed by GAS1: the gas the reading needs is a fixed column $\Sigma_{\rm gas} = a_0/(\pi G x_b^2) = 6.1627/7.4477\,{\rm M_\odot\,pc^{-2}}$, and putting it into $g_{\rm bar}$ moves the outer amplitude to $0.0392 \pm 0.0069/0.0324 \pm 0.0057$ against the 0.891 line. The reading destroys the fit it was invoked to save. The formation reading of the amount scores five of five on FINAL-15's frozen ledger (four of five exactly as frozen) against one of five for the force reading, and the decisive measurement is named with its pre-registered number: a stacked thermal Sunyaev–Zel'dovich measurement on 17,335 host-free KiDS lenses, sensitive to 1.0 % of the required column at 2.6 Mpc. Finally the framework says why $\kappa = \frac{1}{2}$ resisted nine sectors and four dedicated routes: the formation physics depends only on a_0/g, so $G_\lambda : (a_0, g) \mapsto (\lambda a_0, \lambda g)$ is an exact symmetry and κ is its only charged quantity. κ is a flat direction, and the value $\frac{1}{2}$ is a convention the formation reading forces, not a dynamical output.

1 The claim, stated as a claim

There is no dark-matter particle, and there is no second halo. The dark mass of a galaxy is the clock sector's own gravitating stress-energy: a conserved cold charge that sits in force balance around the baryons, whose equilibrium is the law

$$g_{\rm obs}^2 - g_{\rm bar}^2 = a_0\, g_{\rm bar}, \qquad a_0 = \tfrac{1}{2} c\sqrt{G\rho_\Lambda}, \tag{1}$$

set when the system formed and conserved afterwards. Light and matter both feel it; it is not a fifth force and there is no external field effect on real mass. The claim is stated as a claim and not hedged toward ΛCDM: the non-baryonic gravitating density is the sector's own,

$$\rho_c = \nabla\cdot\mathbf{P} + n_d, \qquad n_d = \mu_K^2(Q-1)/K, \tag{2}$$

one conserved charge counted once [C: QX17 K5, FB1, SHP1]. Where the clock free-falls with the expansion that charge is cold dust and is the cold component of cosmology; where the clock stands still around baryons its own equations give the law. Nothing in this paper introduces a particle.

Two footings are carried throughout and **never pooled**: canonical $a_0 = 9.3603 \times 10^{-11}$ and alt $1.1312 \times 10^{-10}\,{\rm m\,s^{-2}}$. Pairs of numbers are canonical/alt unless a row says otherwise. Status tags are [C] certified in a lane of the record, [H] hand algebra checked symbolically, [M] measured, [X] the equations do not give this. None of these lanes is registered in the closure record.

2 The law as a measurement

The statistic is SHP1's and it is built so that an offset is dex in $g_{\rm obs}$ at fixed $g_{\rm bar}$:

$$X = M_C(<r)/M_b(<r) = g_{\rm obs}/g_{\rm bar} - 1, \qquad Y = \log_{10}(1+X), \qquad x = r/r_M, \quad r_M = \sqrt{GM_b/a_0}. \tag{3}$$

The law's prediction is $X(x) = \frac{1}{2}(\sqrt{1+4x^2}-1)$: $\simeq x^2$ inside r_M (density $\sim 1/r$), $\simeq x$ outside it (density $\sim 1/r^2$, an isothermal halo with $\sigma^2 = \frac{1}{2}\sqrt{GM_b a_0}$). In the strong field the column is universal, $\rho_c r = a_0/(2\pi G)$, i.e. $a_0/2\pi G = 106.9/129.1\,{\rm M}_\odot\,{\rm pc}^{-2}$ on the two footings [C: SHP1; H: EQUATIONS].

SPARC, certified accuracy. SPARC has 2700 points in 149 galaxies; the reduction is reproduced exactly (175 files, 2700 points against Lelli+2017's 2693, 149 galaxies against 153). The worst bin median is within 0.057 dex canonical (0.021 alt) for x from 0.18 to 10, against a line of 0.10. Independently, form B passes SPARC rotation curves at rms 0.0563 (line 0.0660) canonical and 0.0558 alt, with no galaxy truncated inside its last measured point (median truncation radius 186 kpc) [C: SHP1, PD1].

KiDS, host-free. On B21's isolated lenses the data have the law's shape and sit above it: $1.40\pm0.03/1.28\pm0.03$ times the law, roughly flat from $x \sim 15$ to ~ 200. On the host-free subset with every catalogued neighbour carrying the same share, form B **passes** its frozen gate at the $1.6\,R_h$ veto for the share $X = 59.49/59.50\,M_b$; the measured outer amplitude there is $1.4002 \pm 0.2211/1.2740 \pm 0.2012$ of the law, against a model of 1.003 canonical / 0.847 alt and **The pre-registered acceptance line is 0.891** [C: ONE1, KO1 E7; reproduced independently by GAS1 to $+0.0002$, 0.00σ]. The pass does not separate form B from the law with no end, which also passes.

That is the certified accuracy of the law: 0.057/0.021 dex on 2700 rotation-curve points, and a pass on host-free KiDS lenses whose error bar is $\pm0.22/\pm0.20$ of the law's level. Both are quoted, not rounded.

3 Form B, in six rules, each tagged

1. **Fields [C].** The metric, one clock field τ that picks a slicing (unit normal n, rate Q), and a polarization P on its leaves ($n\cdot P = 0$). Matter couples to the metric only; baryons fall on geodesics. The action is QX17's (A), certified (label CERTIFIED).
2. **One conserved charge [C].** The clock equation is a conserved current, $\nabla_m[n^m\rho_Q - \lambda P^m/Q] = 0$. The universe carries 5.41 units of that charge per unit of baryons (Ω_c/Ω_b, fitted to the acoustic peaks). It is not matter made of particles; it is a field component with $w \approx 0$.
3. **Falling clock [C].** Where the clock free-falls with the expansion the charge is cold dust. On the background $n_d a^3 = {\rm const}$, $w = w_0/(w_0 + a^3)$, $c_s^2 = (Q-1)/Q$, and the Jeans wavenumber is $k_J = \mu_K$ (physical). This is the cold component of cosmology.
4. **Standing clock [C].** Where the clock stands still around baryons, $K\mathbf{P} = (1-\mu)\nabla\Phi$ with $\mu = u/(1+u)$, $u = |\nabla\Phi|/a_0$, so $\nabla\cdot[\mu\nabla\Phi] + \mu_K^2\Phi = K\rho_b$ in any geometry; spherical with $\mu_K = 0$ this is $g_N = g^2/(g+a_0)$ exactly. The cold pull $g_c = a_0 g/(g+a_0) \le a_0$ saturates. The state is stable with gravity coupled around point masses (SCS1) and isolated discs (SCS2). $\kappa = \frac{1}{2}$ fitted.
5. **Counted once [M].** A halo is made of the charge its region brought, and amounts are conserved: the halo is whatever charge the clock carried in (Section 4). A hot system holds 5.41 times the baryons that fell into it. A body that brought no charge has no halo — measured, as the tidal dwarfs and NGC 1052's UDGs sitting at the floor (Section 6).
6. **The joint [X].** How a falling clock becomes a standing one is not derived, and the amount the standing state needs is not available (Section 4).

Rules 1 and 2 are certified or fitted; rules 3–5 are two solutions of one action; rule 6 is the missing equation, and the hot branch of rule 5 is a rule, not a solution of the action.

The state, and what it says the medium is [H, computed]. The static halo is held by the law's own stress $\Pi_{ij} = J(g)\delta_{ij} - g_i g_j/(K(1+u))$ with $d_j\Pi_{ij} = (\nabla\cdot\mathbf{P})g_i$. In the deep field $\Pi_{rr} = -\Pi_{tt}$ exactly, so the radial "dispersion" is *negative*, $\sigma_r^2 = -v_\infty^2/2$ with $v_\infty^4 = GM_b a_0$, while $\sigma_t^2 = +v_\infty^2/2$. A kinetic gas cannot do this; a medium with a shear modulus can. The computed approach to the identity, $(\Pi_{rr} + \Pi_{tt})/\Pi_{tt}$, is 0.96 at $0.1r_M$,

0.20 at $3r_M$, 0.066 at $30r_M$ and 6.6×10^{-3} at $100r_M$. This is a statement about what the medium is, not a derivation of it.

4 The reach–amount lock

The joint was filed in the record as three failures — in space, in the constant, in time. The exact results make it one failure, and a budget rather than a transport problem: **the charge is not there.**

4.1 T4: the conservation cap (exact)

Inside any ball whose boundary free-falls, the net polarization charge is exactly zero, at all times, for any exchange rate, alignment and multipole — and this holds *for any momentum equation*, variational or not, provided the Legendre map is kept. It is Gauss's theorem, not dynamics, so no candidate is exempt from it by changing the dynamics. The two landed records are **T4 GAUSS (exact; SMF1 S0b reproduced)** and **S0b (exact, Gauss)**. The consequence is one line: polarization **cannot be manufactured during infall**. (T4 has no lane folder of its own; it is carried by AKS1's frozen criteria and SMF1's S0b.)

4.2 LFS1: the standing halo is frozen (exact)

For a static clock the polarization is frozen: the leafwise flux is the normal equation's own bookkeeping and adds no charge, so a static halo cannot grow. Its charge changes only by a fraction of order $|\Phi|/c^2$, about 10^{-6}. From T4 and LFS1 together: **the halo's charge equals the dust charge the clock carried in.** The lane's own label is ILL-POSED on both footings, and that is the answer to the question the record had left open.

4.3 AMT1: the budget, and the demand against it

The dust charge available is the cosmic share, $\mathbf{5.41\,M_b}$; what actually turns around is $\mathbf{1.06\,M_b}$ turned around by $z = 0$ ($1.40\,M_b$ settled, VSH1), because with the law on, the clock's own self-gravity is cancelled below $1/\mu_K$ and only the baryons pull it in. The law demands $M_C(< r) = M_b X(x)$. At FLD1's actual KiDS window radii, for $\log M_b = 10.45$ ($r_M = 6.479\,\mathrm{kpc}$ canonical):

r (kpc)	$x = r/r_M$	demand X $[M_b]$	× budget	where the radius comes from
10	1.543	1.12	0.21	SPARC radii
30	4.630	4.16	0.77	COL1 Part 0
38.2	**5.889**	**5.41**	**1.00**	**the whole cosmic budget is spent here**
139	21.45	20.96	3.87	FLD1 $R_{\rm fold}(z = 0)$
272	**41.98**	**41.48**	**7.67**	**FLD1 KiDS inner window edge**
389	60.00	59.50	11.00	ONE1 end radius, $X_s = 59.5$
637	**98.32**	**97.82**	**18.08**	**FLD1 KiDS middle window edge**

The shortfall is mass-independent and footing-independent: $X_s/5.41 = 10.99/11.01/10.99/11.01$ across the four KiDS masses (alt: 12.09 to 12.11), because the windows scale with r_M. The budget is spent at 38.2 kpc canonical and 34.7 kpc alt for $\log M_b = 10.45$, and at 72/65 kpc for $\log M_b = 11$. The share ONE1 needs, $X_s = 59.5\,M_b$, is 11.00 times the entire budget, 56 times the delivered $1.06\,M_b$, and 42.5 times VSH1's settled $1.40\,M_b$.

And the data reject the budget directly, not through the fit: ONE1's own brackets reject the cosmic share $X = 5.41$ at $+4.85/+4.93\sigma$ (and $X = 10.82$ at $+4.34/+4.53$). So the gap is measured, and whatever pays it has to move a 5σ-measured gap at the radii FLD1 names.

4.4 GAS1: the missing-baryon flank, closed

The one honest reconciliation was that $g_{\rm bar}$ at KiDS radii is missing hot gas. The gas the reading needs is a **fixed surface density**, constant in radius out to 2.6 Mpc and independent of the galaxy:

$$\Sigma_{\rm gas} = \frac{a_0}{\pi G x_b^2}, \qquad x_b^2 = R_b(1 + R_b) = 5.41 \times 6.41 = 34.6781,$$

i.e. $1.287355 \times 10^{-2}\,\mathrm{kg\,m^{-2}} = 6.1627\,\mathrm{M_\odot\,pc^{-2}}$ canonical and 7.4477 alt. Graded against every gas constraint on disk: against all baryons in the same sphere it is over by 284–574; against B21's own hot gas over by 328–3750; against GO1's measured group gas, +8.0 to +16.3σ; against ρ_b (G5) over by 1.38/1.67, the one gate that did not fire. And the gate that closes the loop, G4: with that gas in $g_{\rm bar}$ the outer amplitude sits at $0.0392 \pm 0.0069/0.0324 \pm 0.0057$ of the law, 36/41 times below the law and more than 100σ under the 0.891 line. **MISSING BARYONS EXCLUDED [AMOUNT AND AMPLITUDE]**, the same label on both footings. What survives of the baryon reading is narrower: the column the *cluster* rows need is within a factor ~ 1 of what X-COP measures (ratio 0.37 to 1.26), so it stays open at cluster and group scale. The galaxy leg is closed.

4.5 Two kills that were run so the lock could not be dodged

The anisotropic stress cannot pay the amount. The natural escape is that the stress carries the lensing and the mass need not exist. The active (Tolman) source of a static spherical configuration is $\rho + (p_r + 2p_t)/c^2$, and with $p_r = -p_t$ that is $\rho + \Pi_{tt}/c^2$, so the fractional correction is σ_t^2/c^2: $1.96 \times 10^{-7}/2.16 \times 10^{-7}$ at $\log M_b = 11$, against a shortfall of a factor 5 to 18. The lensing signal is mass.

The elastic reading does not evade M5. $\sigma_r^2 < 0$ requires a shear modulus, i.e. a solid. M5's theorem has three hypotheses: local, momentum-conserving, normally self-gravitating. A solid with a local constitutive relation is the first two, so M5 bites unless the self-gravity is not normal. That is ELA1's gate, and ELA1 returns ELASTIC SOLID FAILS AT S0 (+S2, B1) on both footings: the deformation fires, the energy $W = J$ fires, and the Bullet fires.

5 The closure ledger: every lane, with its verdict

Every label below is a lane's own first line, quoted; where a lane and any summary differ, the lane is right. Post hoc amendments are named, and the frozen one is given first.

5.1 The week's own lanes (this session and its siblings)

Lane	Question	Verdict (canonical / alt)
FINAL-1	assemble the theory the record supports, in seven pieces	THEORY WRITTEN / THEORY WRITTEN
FINAL-2	which formation event fixes the halo's amount	(e) BEST-RANKED / NO POWER (both)
FINAL-4	the structure-growth history the formation reading gives	DERIVED HISTORY EXCLUDED (9916/9009); TILT POST HOC, NOT EXCLUDED
FINAL-6	is κ an output of the formation reading?	KAPPA NOT DERIVED (FITTED) + FORMATION REFUTED (per-galaxy)
FINAL-15	the formation reading, written as a theory and graded	FORMATION LEDGER HOLDS / HOLDS IN PATTERN, NOT IN SIZE
FINAL-16	κ from the real cosmic density	KAPPA NOT DERIVED (FITTED) – FORM OUTPUT
FINAL-17	κ from the cosmic capacity	KAPPA NOT DERIVED (FITTED) – CAPACITY DERIVED
FINAL-18	κ from a matching condition at the fold	KAPPA NOT DERIVED (FITTED) – FOLD MATCHING FALSIFIED, NO SOLUTION IN THE VALID RANGE

Lane	Question	Verdict (canonical / alt)
FINAL-19	κ from the deep-field column	kappa = 1/2 stays FITTED (6.0±1.4 column; $\sqrt{2\pi}$ = 2.5066 at 22.1/27.6σ)
FINAL5	the decisive observation	runnable today with archival data (stacked tSZ, 17,335 lenses)
HY4-1 GAS1	the missing-baryon flank	MISSING BARYONS EXCLUDED [AMOUNT AND AMPLITUDE] (both)
ELA1	the medium as an isotropic elastic solid	ELASTIC SOLID FAILS AT S0 (+S2, B1) (both)
PIK1	the two-variable dust term $L_K(Q, \Pi)$	NO VALUE / NO VALUE
VCC1	sourcing the clock's charge	EXCLUDED / EXCLUDED
CLS2	what the cluster's non-baryonic mass is	NO CANDIDATE / NO CANDIDATE
FINAL-14	the amount arithmetic, in Lean	six items, exit 0, zero sorry

FINAL-2's honest detail: no epoch among six candidates reproduces the measured pattern; the end of active star formation is the least wrong and beats the no-epoch floor by 0.056 dex, inside the class errors. FINAL-4's: the formation reading *derives* FRZ1's baseline, $\Delta\chi^2 = 9916.246/9008.690$ against a gate of 4, and every one of the twelve bins fails on both footings; the three post hoc tilt cases pass the same gate, so the gates confirm the tilt rather than excluding it — a gate you were fitted to cannot exclude you, and the test that can is frozen. FINAL-15's specification defect D1 is disclosed: exactly as the criteria were frozen, without the correction, the label is FAILS / FAILS.

5.2 What the fable52 push closed (the class, on every exit)

Statement	Verdict	Lanes
Keep the cold component out of galaxies by any action-based switch	closed on every exit	SWT1, EMV1, GAL1, KO1, KO1R, SNP1, HE1
Hold the galaxy's share outside a wall, statically	FAILS / FAILS	HE1
One dust constant for CMB, clusters and galaxies (quadratic)	NO VALUE / NO VALUE	MK1
An exponential dust term (one sound speed)	NO VALUE / NO VALUE	MK2
A clock that starts as dust settles into the law's halo	BREAKS DOWN (folds in 0.06–0.15 Gyr)	COL1
The folded region reaches where the law is seen	DOES NOT (1.3–5.4× short)	FLD1
The law keyed on the clock's acceleration relative to baryons	NOT WELL-DEFINED (4th order)	REL1
The law keyed on the clock's expansion	BREAKS AT TURNAROUND	TK1
The leafwise flux supplies the halo's charge	ILL-POSED: a static halo cannot grow	LFS1

Statement	Verdict	Lanes
Polarization charge inside a free-fall-bounded ball	exactly 0, any time, any momentum equation	T4 (SMF1 S0b, AKS1)
a_0 as a contact opacity with neutral gas	KILLED (K2 PROVISIONAL)	NBC1
The clock is the dark-energy field	ALREADY IN THE RECORD AS AC2s	KDE1
Baryons + law build late structure with a warm component	FALLS SHORT	GRW1
A held edge layer of real charge	FAILS AT E3 (19–38× short)	EDG1
Parent actions: single-valued carrier / wave phase / smooth frame	EMPTY, STRUCTURE FAILS, STRUCTURE FAILS	PAR1, WV1, SMF1
Any local, momentum-conserving, normally self-gravitating medium	NO-GO, a theorem	M5
Every medium class the record can name	a failing verdict for each	MEDIUM_DESIGN
The data, asked with no theory, return a single rule	NO RULE AMONG THE CANDIDATES	DD1

5.3 What stands under form B

Statement	Verdict	Lanes
The law's halo is the clock sector's own gravitating density, on one conserved charge	certified	QX17, FB1
The static state holds the law's shape (force balance, any geometry)	checked, exact	FB1
That state is stable with gravity coupled	STABLE, point masses and isolated discs	SCS1, SCS2
The law's shape from raw rotation-curve data	0.057 / 0.021 dex	SHP1, DD1
Form B on host-free KiDS lenses	PASSES at the $1.6\,R_h$ veto	ONE1
Form B's cosmology rows	ROWS HOLD, inherited halo list	CO1
Form B on groups, clusters, the Bullet	not falsified (one-sided), never a pass	RB1

5.4 The Lean record

FINAL-14 certifies the amount arithmetic as named theorems with 'lean' exit code 0, zero errors, zero warnings, zero sorry, zero sorryAx: X is monotone, $X(x_b R) = R$ exactly (the identity $1 + 4R(1 + R) = (2R + 1)^2$), the break-even window $x_b(541/100) \in (5.8887, 5.8889)$, the demand-over-budget ratios 3.87 / 7.67 / 11.00 / 18.08, and the gas column 6.1627 / 7.4477. FINAL-12 compiles 30 theorems with the same axiom set (subseteq propext, Classical.choice, Quot.sound) and no sorry; FINAL-16's four theorems carry the same axioms and its compiled source is byte-identical to the committed blob (md5 db7a54eb96c684cef667c83f87a89a4b); FINAL-18's Lean proves 'fold_match_unique' and 'fold_match_exists'. fable52's own Lean files hold 92 theorems, each with

a false twin that fails. Where a lane's Lean is not clean the record says so: FINAL-10 and FINAL-12 were recorded as FAIL with sorryAx disclosed.

6 The formation reading, and its score

The amount is an initial condition, not an advected quantity. The reading states that the non-baryonic mass is real and conserved and that its amount is set by the law when the system formed; the alternative is that the law is a force every system obeys now. For a settled, closed, complete system the two are the same function, which is why SPARC cannot decide between them. They come apart only where history is not the identity, and there the formation reading predicts three things the force reading cannot produce at any setting.

1. **Quantisation of the deficit.** Recycled material sits at the floor, $d = -Y_{\rm law}$, $X = 0$, Newton — not partway. Measured: tidal dwarfs -0.687 against a floor of -0.755 (0.07 dex from it, 6.0 se below the law); NGC 1052's UDGs -0.843 against -0.855 (0.01 dex, 3.4 se).
2. **Bipolarity with two unrelated caps.** Deficits can only come from charge stripping or late accretion and cannot pass the floor; excesses can only come from baryon loss and cannot pass the ceiling $d \leq \log_{10}[(1+u)/(1+X_L(u))]$, which is the law's own certified bound $g_c \leq a_0$. The measured 1.435 dex bipolar spread, with classes at the floor on one side and above the ceiling on the other, is the reading's ordinary expectation and is forbidden by the force reading at 20–24 se.
3. **Class-level coherence with within-class incoherence.** The amount is a per-system constant written on the assembly hypersurface, so classes separate; what varies is an integral over history, so no present-day proxy tracks it inside a class. The within-class null is a prediction of the reading, not a hole in it.

The score. FINAL-15 grades five frozen gates against DD1's frozen 626-system, 16-class table. Exactly as frozen: F1 FAILS (defect D1: a 3 se line was written as 5 se, unsatisfiable at $n = 2$), F2, F3, F4, F5 HOLD — four of five. With D1 corrected to the line used everywhere else in DD1's criteria: all five hold. So the ledger is **five of five corrected, four of five frozen**, on canonical; on alt it holds in pattern and not in size (F3's SPARC residual is -0.023 ± 0.012, the alt footing's own anchor, and the hosted excesses are smaller by 0.04 dex). The independent, separately-coded regrade HIS1R reaches the same label, with 145 of 148 compared numbers agreeing to 5×10^{-12} and the three that differ moving no label; it discloses that it was not fully blind. HIS1's own comparison on the same table: the formation reading scores 4 of 5 and the force reading 1 of 5. FINAL-2's contribution is a negative that the reading survives: no epoch candidate reproduces the pattern, which is consistent with the amount being an integral over history rather than a dated event.

7 The decisive measurement, with its pre-registered number

One number carries the whole shortfall: the share $X_s = 59.5\, M_b$ that ONE1 needs to pass KiDS host-free, against a cosmic budget of 5.41. The two are reconciled only if a galaxy's halo holds about 11 baryons' worth of cosmic share sitting at the radii KiDS measures. The record's own words put the flank first: "with B21's own hot-gas estimate in $g_{\rm bar}$, its published outer points drop from 1.50 to 0.47 of the law. Nothing on disk measures that gas."

The measurement. Not a new instrument and not a new exposure: a **stacked thermal Sunyaev–Zel'dovich measurement in the ACT DR6.02 needlet-ILC Compton-y map already on this disk**, centred on the **17,335 KiDS host-free isolated L$\star$ lenses** the record has already built, run through GO1's existing Part A pipeline, which already stacks KiDS shear and this y map with randoms, jackknife errors, a gNFW beam correction, two CIB treatments and three nulls. It is decisive because y is linear in the very quantity the budget fixes:

$$y(b) = \frac{\sigma_T}{m_e c^2} k T_e\, N_e(b), \qquad N_e(b) = \int n_e\, dl = \Sigma_{\rm proj}(b)/(\mu_e m_p),$$

so N_e *is* the required surface density; there is no clumping factor and no $\langle n^2 \rangle$ bias, and in the budget-forced ($\rho \sim R^{-1}$) reading the plateau is flat in b. Sharper still, the fixed column is exactly the statement $g_{\rm bar}(R) = \pi G \Sigma_{\rm gas} = a_0/x_b^2 = 2.699196 \times 10^{-12}\,{\rm m\,s^{-2}}$ canonical (3.262001×10^{-12} alt) — a parameter-free flat baryonic-acceleration plateau from ~ 66 kpc to 2.6 Mpc.

The pre-registered number. A straight stack of 17,335 lenses measures the projected gas column to 1.0 % of the requirement at 2.6 Mpc ($\sigma(\Sigma)/\Sigma_{\rm gas} = 0.0101$, S/N 99 if the column is real) and to 8.8 % at 0.3 Mpc

(beam-limited, S/N 11). If the column is real this is a $\sim 100\sigma$ detection; if it is not, the reading is dead by the same factor. There is no outcome in between. The amplitude gate that any gas in $g_{\rm bar}$ must pass is the record's own: **The pre-registered acceptance line is 0.891**, against which GAS1's budget-fixing column gives $0.0392 \pm 0.0069/0.0324 \pm 0.0057$.

The data already lean that way, but are not yet inside the kill region. GO1's ACT-y stack of 102 eRASS1 groups ($\sim 10\times$ an L$\star$ halo, so expected to carry more gas, not less) supplies $f_{\rm col} = 0.187 \pm 0.102$ of the requirement at 1.87 Mpc, a deficit of $+8.0\sigma$, reaching $+13.3\sigma$ at 7.46 Mpc. Against the only Mpc-scale electron densities in the repo the required column is over by $2.5\times$ ($+5.7\sigma$) at 0.9 Mpc and $10.5\times$ ($+26\sigma$) at 1.9 Mpc. Both facts are the reason to run the stack rather than declare victory: 0.19 ± 0.10 sits only $+0.9\sigma$ above the 0.10 line, and those are groups, not isolated galaxies.

The dilemma the measurement resolves. Flat column $\Rightarrow \rho \sim R^{-1} \Rightarrow 2483\times$ the cosmic mean baryon density at 0.3 Mpc and a gas mass inside 1 Mpc of $101\times$ the total baryons an L$\star$ halo can hold: impossible. A normal halo shape $\rho \sim R^{-2}$ matched at 1 Mpc gives $0.77\times$ the requirement at 2.6 Mpc and over-supplies at 0.3 Mpc, which drives the outer points to 0.03–0.04 of the law: short. No third shape exists. Both branches are excluded by measurements already on disk.

8 The framework: the scaling symmetry, the flat direction, and the forced convention

Nine sectors (NPJ1) and four dedicated routes (FINAL-6, FINAL-16, FINAL-17, FINAL-19) failed to derive $\kappa = \frac{1}{2}$, and one route (FINAL-18) falsified the only condition that was not a flat direction. The reason is structural, not numerical.

8.1 The exact symmetry

The formation physics depends only on the ratio a_0/g. Therefore the map

$$G_\lambda : \ (a_0, g) \ \mapsto \ (\lambda a_0, \ \lambda g)$$

is an **exact symmetry** of the formation equations, and κ is the only quantity charged under it. So κ is a flat direction: no dynamics can determine it, any more than dynamics can determine the value of the metre. FINAL-16 puts it in one line — the route "derives only the FORM $a_0 = \kappa\, c\sqrt{G\rho_\Lambda}$."

8.2 The routes, and the freedom each one carries

Each route that *lands* on $\frac{1}{2}$ is circular by construction, and each says so in its own output.

Route	What it fixes	Freedom it carries	Result
FINAL-16	the form, and $\rho_* = \rho_\Lambda$ as the unique epoch-independent cosmic density	0.319 dex across three named epochs, against a 0.0411 dex allowance	form OUTPUT, κ NO
FINAL-17	the capacity $c\sqrt{G\rho}$	0.798 dex across five cosmic accelerations, $19.4\times$ the allowance	capacity DERIVED, κ NO
FINAL-19	the deep-field column	the halo can sustain 6.0 ± 1.4 / 7.3 ± 1.6 times the universal column; saturation inverts to $\kappa = \sqrt{2\pi} = 2.5066$ at $22.1/27.6\sigma$	κ stays FITTED
FINAL-18	a matching condition at the fold	*not* a flat direction ($176\times$ more constraint than freedom), and it fails: 1.24–2.23 dex mismatch, 1.86 dex drift with mass, 0.37 dex with epoch, $\kappa_M = 1.19 \times 10^{-3}$	FOLD MATCHING FALSIFIED

FINAL-6 adds the diagnosis of the circularity: $\kappa = 1/(2\sqrt{\beta})$ with $\beta \equiv \rho_m(z_f)/\rho_\Lambda$, so $\kappa = \frac{1}{2}$ *is* the statement $\beta = 1$, freeze-out exactly at matter–Λ equality. The lane convicts this in its own printed output, and FINAL-6's

second half refutes the per-galaxy formation route outright: both halo-scale routes predict $a_0 \propto M^{1/3}$, which the universality of the RAR excludes.

FINAL-19 is worth quoting for what it says about the vacuum. The halo's deployable anisotropic stress is $|\Pi|_c = a_0^2/(2\pi G)$, four times the pressure scale $a_0^2/(8\pi G)$; demanding that the vacuum saturate it inverts to $\kappa = \sqrt{2\pi}$. To make $\kappa = \frac{1}{2}$ come out, the vacuum's own energy density must be divided by Einstein's 8π before it can act as that stress — and that factor is exactly the content of the identity the lane was told not to assume.

8.3 What is actually derived, and what it costs

Universality kills the mass monomial; conservation kills the epoch-bearing densities $\rho_{\rm crit}(z_f)$ and $\rho_m(z_f)$. The unique epoch-independent cosmic density is ρ_Λ. Hence $\rho_* = \rho_\Lambda$ and $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ with $\kappa = \frac{1}{2}$ is forced — as a **convention-fixing**, not as a dynamical output. That is exactly why it survived nine sectors and four routes: they were looking in the dynamics for something the dynamics is symmetric under.

The cost is stated rather than hidden. $\kappa = \frac{1}{2}$ is a gauge choice, not a prediction. The framework explains *why* the coincidence $a_0 \approx cH_0/6$ holds — it must, because $c\sqrt{G\rho_\Lambda}$ is the only dimensionful combination and its coefficient is conventional — and it does **not** predict the numerical relation beyond the convention. The framework absorbs the coincidence rather than predicting it. FINAL-16 also prints the external value that breaks the footing tie, $\kappa_{\rm ext} = 0.6414 \pm 0.0107\,({\rm rand}) \pm 0.1283\,({\rm sys})$.

9 What survives, what is open, and what would decide it

What survives. The a_0 relation, the shape $X(x)$, and the deep-field column $a_0/2\pi G$: certified. The halo as a real, gravitating, conserved non-baryonic mass: certified by the data, and by DD1's "nothing below zero among 626". Its radial shape when settled: certified. The framework's form $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ with ρ_Λ selected: derived. The stability of the standing state with gravity coupled: certified.

What is open, honestly.

- The amount, on the galaxy leg: an 11.0× budget gap that is measured, not inferred. Clusters (+0.5 dex) have no candidate (CLS2). Hosted dwarfs (+0.5 to +0.6 dex) fail (RB2).
- What the cold component *is*: every candidate has a failing verdict. The joint is not derived.
- A disc in an external field: no verdict (SCS2, SCS3). Lensing by galaxy type: no verdict (KO2).
- Growth: there is no derived growth history. The required one is being computed and is currently empty (SRH1 WINDOW EMPTY); FINAL-4 excluded the derived history and left the post hoc tilt not excluded.
- One engineered function, a kinetic function with a narrow interval of changed compressibility (DT2), and form B as a set of rules. The rules are complete as a predictive theory; there is no equation of motion that produces them.

What would decide it, in order. (i) The stacked tSZ measurement of Section 7: it is runnable on archival data and it returns a yes or a no on the amount at the radii that matter. (ii) Whether $59.5\,M_b$ is a requirement or a census error — the first thing to decide is not a mechanism. (iii) A formation-set share: the amount as an output of the collapse rather than an input rule, with the epoch fixed (FINAL-2 says no dated event works, so the conserved quantity, not the date, is the object).

10 Our own estimates that were wrong

Registered before GAS1 ran: *"GAS1 will find the census closes at least half of the 11.0×."* Wrong. GAS1 fired four of five gates against it and the amplitude gate killed it outright, 36–41× below the law's level. The bet is recorded as lost, not quietly dropped. FINAL-15's frozen clause F1(b) was unsatisfiable as written (defect D1); both labels are printed and nothing else in the lane changes between them. FINAL-4's tilt was fitted to gates it was later graded on, and the lane says so: a gate you were fitted to cannot exclude you. FINAL-14 flags that its brief's $G = 6.6743 \times 10^{-11}$ and the quoted 6.1627/7.4477 come from $G = 6.674 \times 10^{-11}$; both are certified and agree to three decimals.

11 What is from memory

The cooling function $\Lambda(T, Z) = 3.0 \times 10^{-23}$ used for FINAL5's X-ray fluxes (it moves X-ray fluxes by up to ± 0.5 dex and nothing else). The local column budget $21 \pm 5\,\mathrm{M}_\odot\,\mathrm{pc}^{-2}$ quoted in NBC1. The Chabrier-like mass-to-light ratio behind HIS1's early-type hold. Everything else quoted here is computed in a lane or read from a file on disk.

Method

Every number is quoted from the lane named beside it, and PAPER74_audit.py re-checks each quoted literal against that file in the working tree, at HEAD and on origin/main, and recomputes the paper's own arithmetic; '–mutate' changes one digit of each source literal and requires every mutated literal to fail. Lanes are quarantined: they adopt nothing, relabel nothing and register nothing, and where a lane and this paper differ the lane is right. Footings are never pooled. The cold component is a gravitating field component and is never a particle.